

**ASSIGNMENT 04****Due date: Sunday, 31 August 2025**

Total Marks: 40

**ONLY FOR YEAR MODULE**

This assignment covers **chapter 2 & 3 of the prescribed book as well as the study guide**, it is based on Study Units 2.1, 2.2, 2.3, 3.1, 3.2 & 3.3 of Chapters 2 & 3 of HCNote that this assignment will close on **29 August 2026 at 9:00 PM**

**Question 1: 10 Marks**

Let  $\vec{u} = \langle 1, 1, 0 \rangle$  and  $\vec{v} = \langle 0, 1, 1 \rangle$ . A third vector  $\vec{w}$  has a magnitude of  $\sqrt{12}$  and is in the direction of the vector projection of  $\vec{a} = \langle 2, 2, 1 \rangle$  onto  $\vec{b} = \langle 1, 0, -2 \rangle$ .

- Find the vector  $\vec{w}$ .
- The vectors  $\vec{u}$ ,  $\vec{v}$ , and  $\vec{w}$  are used to define two planes:
  - Plane  $\Pi_1$  contains the origin and is spanned by the vectors  $\vec{u}$  and  $\vec{v}$ .
  - Plane  $\Pi_2$  has  $\vec{w}$  as its normal vector and contains the point  $P(1, 1, 1)$ .

Find the shortest distance between the line of intersection of  $\Pi_1$  and  $\Pi_2$ , and the line  $L$  which passes through the point  $Q(0, 5, 0)$  and is parallel to the vector  $\vec{u} \times \vec{v}$ .

**Question 2: 10 Marks**

Consider the set of all points  $P(x, y, z)$  in  $\mathbb{R}^3$  such that the vector from the origin to  $P$  can be written as  $\vec{p} = \vec{a} + (\vec{a} \times \vec{b})$ , where  $\vec{a}$  is a unit vector that can point in any direction, and  $\vec{b} = \langle 0, 0, 1 \rangle$ .

- Prove that all such points  $P$  lie on the surface of a sphere. Find the center and radius of this sphere. (Hint: Let  $\vec{a} = \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle$  in spherical coordinates).
- Show that the volume of the parallelepiped formed by the vectors  $\vec{a}$ ,  $\vec{b}$ , and  $\vec{p}$  is independent of the choice of  $\vec{a}$ .
- Find the specific direction(s) of the unit vector  $\vec{a}$  for which the vector  $\vec{p}$  is orthogonal to the vector  $\vec{c} = \langle 1, 1, 1 \rangle$ .

**Question 3: 10 Marks**

Let  $z_1, z_2, z_3$  be the three cube roots of  $w = 4\sqrt{2}(-1 + i)$ . These roots correspond to three points  $P_1, P_2, P_3$  in the complex plane.

- a) Find  $z_1, z_2, z_3$  in polar and Cartesian form.
- b) These points are treated as vertices of a triangle in the plane. A fourth point  $P_4$  is chosen such that  $P_1P_2P_3P_4$  forms a rhombus. Find the possible coordinates of  $P_4$ .
- c) The points  $P_1, P_2, P_3$  are now considered as vectors in  $\mathbb{R}^3$  with a z-component of 1, i.e.,  $\vec{v}_k = \langle \text{Re}(z_k), \text{Im}(z_k), 1 \rangle$ . Find the equation of the plane containing these three points, and find the area of the triangle they form in  $\mathbb{R}^3$ .

#### Question 4: 10 Marks

Let  $z_1, z_2, z_3$  be the three cube roots of  $w = 4\sqrt{2}(-1 + i)$ . These roots correspond to three points  $P_1, P_2, P_3$  in the complex plane.

- a) Find  $z_1, z_2, z_3$  in polar and Cartesian form.
- b) These points are treated as vertices of a triangle in the plane. A fourth point  $P_4$  is chosen such that  $P_1P_2P_3P_4$  forms a rhombus. Find the possible coordinates of  $P_4$ .
- c) The points  $P_1, P_2, P_3$  are now considered as vectors in  $\mathbb{R}^3$  with a z-component of 1, i.e.,  $\vec{v}_k = \langle \text{Re}(z_k), \text{Im}(z_k), 1 \rangle$ . Find the equation of the plane containing these three points, and find the area of the triangle they form in  $\mathbb{R}^3$ .